

3.4 Sorting and Pagination in Spring Data Jpa

- Sorting with method queries:

- OrderBy:

- @Repository

- public interface EmployeeRepository extends JpaRepository<Employee, Long> {
List<Employee> findAllByNameAsc();
List<Employee> findAllByNameDesc();
}

- Sorting with the Sort class:

- Sort Parameter In Query Methods:

- @Repository

- public interface EmployeeRepository extends JpaRepository<Employee, Long> {
List<Employee> findByDepartment(String department, Sort sort);
}

- Using the Sort class:

- Sort sort = Sort.by(Sort.Direction.ASC, sortField);
Sort sort = Sort.by(Sort.Order.asc("name"), Sort.Order.desc("salary"));

- Key Concepts of Pagination:

- Page:

- ▶ A Page is an **interface** that **represents** the **result** of a **paginated query**.
 - ▶ It contains:
 - The data of the current page (actual content)
 - The total number of elements
 - The total number of pages
 - Information about the current page (page num, ele count in cur page)
 - ▶ In simple terms, **Page** is **what you receive** after **pagination** is **applied**.

- Pageable:

- ▶ Pageable is an **interface** that **represents pagination instructions**.
 - ▶ It defines:
 - Which page to fetch
 - How many records per page
 - How the data should be sorted
 - ▶ It does **not perform pagination** itself—it only **carries pagination information**.

- PageRequest:

- ▶ PageRequest is a **concrete implementation** of the **Pageable interface**.
 - ▶ It provides **factory methods** to create a **Pageable object** that holds **pagination** and **sorting details**.
 - ▶ **Spring Data JPA** then uses this **Pageable object** to execute **paginated queries**.

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- Using Pageable:

```
public interface UserRepository extends JpaRepository<User, Long> {  
    Page<User> findAll(Pageable pageable);  
    Page<User> findByLastName(String lastName, Pageable pageable);  
}
```

- **Creating Pageable instance:**

```
Pageable pageable = PageRequest.of(pageNumber, size,  
    Sort.by("age").ascending());
```